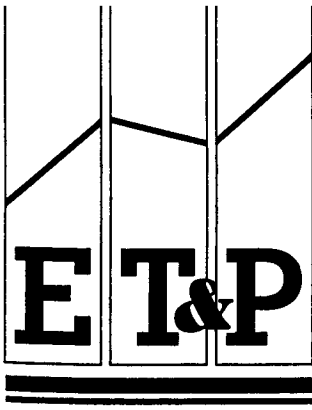




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A Dynamic Model of Entrepreneurial Learning

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We model entrepreneurial learning as a calibrated algorithm of an iterated choice problem in which entrepreneurs learn by updating a subjective stock of knowledge accumulated on the basis of past experiences. Specifically, we argue that entrepreneurs repeat only those choices that appear most promising and discard the ones that resulted in failure. The contribution of the paper is twofold. First, we provide a structural model of entrepreneurial learning in which failure is as informative—though clearly not as desirable—as success. Second, to complement standard economic models in which agents are rational, we allow our entrepreneurs to have myopic foresight. Our entrepreneurs process information, make mistakes, update their decisional algorithms and, possibly, through this struggle, improve their performance.

*"Success is the ability to go from failure to failure without losing your enthusiasm."
Winston Churchill*

Kirzner (1973, 1979) describes entrepreneurship as the outcome of superior alertness exhibited by selected individuals in the population. But, once an opportunity is recognized, how do entrepreneurs go about its exploitation? Knight (1921) characterizes the entrepreneur as someone able to cope with uncertainty. But how do entrepreneurs decide what actions are best suited to cope with uncertainty? To address these questions we build a model of the decisional algorithm followed by individuals faced by an array of choices with uncertain outcomes. That is, we build a model in which entrepreneurs, or potential entrepreneurs, have to recognize desirable opportunities and cope with the uncertainty associated with the payoff of these opportunities. We assume entrepreneurial decisions to be the result of an entrepreneur's ability to process information (knowledge), and of random impulses (instinct or luck). In the long run, it is the knowledge component that determines the entrepreneur's selection of the most appropriate course of action in any specific uncertain environment. In particular, we argue, entrepreneurs learn from successes as well as failures: The combination of positive and negative experiences molds entrepreneurs' knowledge and determines the sequence of their choices. Such a sequence is specific to each entrepreneur and, since individuals can make mistakes and the future is unknown, it is not necessarily optimal.

If entrepreneurial behavior is the result of random impulses (instinct or luck) and of the ability to process information (knowledge), then entrepreneurial learning can be represented as a calibrated algorithm of an iterated choice problem. Since entrepreneurial decisions may result in failures, this iterated choice problem must be formulated as one allowing for the existence of alternative patterns of entrepreneurial decisions, including non-optimal ones. That is, the model must allow for multiple equilibria to exist. Only in this case can we learn how entrepreneurs internalize information from both successes and failures and, ultimately, how they learn.

When making decisions, entrepreneurs choose one of two possible strategies: In the first case they choose actions that replicate or are closely related to the ones they have

already taken, thereby exploiting their pre-existing knowledge. In the second case, entrepreneurs choose new actions that are distinct from the ones they have already taken. In some instances this strategy is justified by previous failures, in others by an incentive to diversify and explore new options. In either case, the decisions of the entrepreneur are a function of two types of knowledge. First, there is specific knowledge about the chosen market. This knowledge involves technical aspects and is product and industry specific. It may be acquired through direct experience or, indirectly, through other means, for example, by hiring people. This type of knowledge requires the explorations of a new course of action every time some kind of innovation is introduced, for example, in the form of an improvement of the product, or because of a new marketing channel (e.g., e-commerce). Second, there is the more general knowledge about "how to be entrepreneurial." This knowledge can be acquired only through learning-by-doing or by direct observation. It is this type of knowledge that marks the difference between an engineer and an entrepreneur.¹ We focus on this second type of knowledge. Specifically, our purpose is to construct a general model of entrepreneurial learning from which to derive testable hypotheses useful for understanding how entrepreneurs develop their craft and, in a sense, what makes them special.²

Knowledge acquired through learning-by-doing takes place when agents choose among alternative actions whose payoffs are uncertain and, as a result, risky. Over time, agents repeat only those choices that appear most promising and discard the ones that resulted in failure. It follows that, because of their outcomes, some actions become a systematic component of the decisional algorithm followed by entrepreneurs and, as a result, possess a self-reinforcing capacity (Minniti & Bygrave 1999, 2000). In an alternative, we may say that these actions exhibit increasing returns to adoption, in the sense that their impact on the decisions of the entrepreneurs is amplified by their repetition.³ We believe this amplified effect to be a fundamental component of the behavior of entrepreneurs, and a key determinant of their ability to achieve successful outcomes. In our model, at every point in time, entrepreneurs are faced by a given number of alternative actions. On the basis of past experiences, entrepreneurs try to maximize profits by making appropriate choices. Often, the actual payoffs from these choices are different from the expected payoffs, in particular because of the possibility of failure. Clearly, entrepreneurs make mistakes. However, they internalize the new available information and update the algorithms upon which they base the calculation of future expected payoffs. Thus, entrepreneurs learn even from failures.

The contribution of the paper is twofold. First, we provide a model of entrepreneurial learning, one in which failure is as informative, though clearly not as desirable, as success. To our knowledge, no such model existed before. Second, in an alternative to standard economic models in which agents are rational, we allow our entrepreneurs to have myopic foresight. We are confident that such a characterization provides a more satisfactory description of the way in which entrepreneurs learn from past experience. Our agents process information, make mistakes, update their decisional algorithms and, possibly, through this struggle, improve their performance.

1. Clearly the two are not mutually exclusive.

2. In this paper, we treat the entrepreneurial unit as a simple homogenous agent. Clearly, we do recognize the complexity and importance for knowledge and learning of alternative organizational structure. Specifically, we subscribe to the argument that "human learning in the context of an organization is very much influenced by the organization, has consequences on the organization, and produces phenomena at the organizational level that go beyond anything we would simply infer by observing learning processes in isolated individuals" (Simon, 1991, p. 126).

3. Our argument is complementary to the idea of positive spillovers developed in the economic literature on human capital accumulation and learning by doing (Jovanovic and Lach, 1989; Young, 1993; Stokey, 1988).

LEARNING AND KNOWLEDGE IN ENTREPRENEURSHIP THEORY

In 1973, Kirzner used the term “entrepreneurship” to identify the element in human action that corresponds to the passage of time, that makes things happen coming that creates change. In Kirzner’s system, “there is present in all human action an element which, although crucial to economizing, maximizing, or efficiency criteria, cannot itself be analyzed in [those] terms” (Kirzner, 1973, p. 31). This entrepreneurial element in human action is the cause of changes at the individual level and in the market, and it is the outcome of superior “alertness” exhibited by selected individuals. Specifically, Kirzner defines such alertness as “the knowledge of where to find market data” (Kirzner, 1973, p. 67). Furthermore, he defines entrepreneurial knowledge as “a rarefied, abstract type of knowledge—the knowledge of where to obtain information (or other resources) and of how to deploy it” (Kirzner, 1979, p. 8). Thus knowledge is the alertness leading to the discovery of opportunities. The entrepreneur did not previously know of the opportunity he acts upon. Nor does anyone else. If he or someone else had known of it, it would already have been seized. But, the entrepreneur’s alertness leads to something previously unimagined—that is, to the discovery of a new way of doing things. This act of discovery transforms the entrepreneur’s knowledge and it is itself a change in the entrepreneur’s stock of knowledge.

Actions following from the initial discovery induce further change in the entrepreneur’s knowledge. Thus, entrepreneurship is a process of learning, and a theory of entrepreneurship requires a theory of learning. Of course, what is learned may be false. Entrepreneurs may fail. But entrepreneurs (like all individuals) learn also from failures. Each entrepreneur organizes past experiences in a set of information that, at any point in time, determines his stock of knowledge. An entrepreneur’s stock of knowledge is molded by his subjective circumstances and his interests determine which elements of his knowledge are relevant to him and his purposes. So, over time, any act of entrepreneurship is a change in the content of the entrepreneur’s knowledge in some area.

Clearly, an entrepreneur’s actions are not independent from one another, thus learning is a process involving repetition and experimentation that increases the entrepreneur’s confidence in certain actions and improves the content of his stock of knowledge. Acquired knowledge generates routines and decisional procedures. Routines are patterns derived from successful solutions to some particular problem (Nelson & Winter, 1982). Also, knowledge has idiosyncratic uses and cannot be used indefinitely without losses. That is, as other assets, it involves specific transaction costs (Williamson, 1975, 1985). These costs limit an entrepreneur’s motivation to search indefinitely for optimal routines. Finally, knowledge is cumulative. What is learned in one period builds upon what was learned in an earlier period. Thus, history matters (Arthur, 1989). An entrepreneur’s previous investments and repertoire of routines constrains his future behavior.

The path-dependent nature of knowledge, the subjectivity of routines, and the possibility of failure, all imply that any satisfactory model of entrepreneurial behavior will have to depart from the standard rational expectation behavior assumed by traditional neoclassical economic models (Frydman, 1982; Bullard, 1994). Also, although very useful for many problems, neoclassical theories of industrial organization (e.g., Tirole, 1989) have little to say about how entrepreneurial knowledge is acquired and how entrepreneurs change and update their information systems. This is so because traditional production function approaches to economic activity only describe a technological relationship between inputs and outputs. In such a relationship, no role exists for entrepreneurial alertness. But in fact, unless the production activity so described is completely contained within a mechanical process, issues of organizational skills and judgments are always involved. Thus, significant improvements in our understanding of entrepreneur-

ship and, more generally of organizations, may come from analyzing how entrepreneurs accumulate and update knowledge—that is, from the study of entrepreneurial learning.

Typically, problems involving fluctuation dynamics with multiple equilibria are modeled by means of non-linear differential equations with Markovian disturbances (Azariadis & Drazen, 1990). Many path-dependent problems, however, follow a pretty general non-linear probability structure. Thus, to derive our results, and to complement standard models, we use a simple example of non-linear path-dependent stochastic process. Such models use a class of dynamic systems characterized by a feedback mechanism that causes some patterns to be self-reinforcing. Often there is a multiplicity of such patterns. As a result, these models are particularly well suited to describe the way in which past experiences contribute to determine an individual's future choices. This is so because in these systems, the early accumulations of small events push the process into one among all possible patterns and eventually lock in the structure. Thus, in an alternative to standard non-linear differential equation models, and following closely Arthur (1995), we present a simple learning algorithm with boundedly rational agents. Recent models of bounded rationality explore actual human behavior and examine its consequences in the context of path dependency (Arthur, 1991, 1994; Chen & White, 1998). Specifically, their ability to accommodate possible suboptimal choices, multiple equilibria, and path dependency, makes these models particularly well suited for the study of real entrepreneurial learning.

SUCCESS, FAILURE AND ENTREPRENEURIAL LEARNING

In general, the learning process can be described as the outcome of a sequence of choices among competing beliefs or actions, whose relative influence over an individual's decisions increases or decreases over time as new experiences take place (Arthur, 1993; Bullard & Duffy, 1999; Dawid, 1996; Riechmann, 1999). Our entrepreneur wishes to maximize the subjective expected discounted value from his choices consistently with his intent. At each time, he chooses one among N possible actions. For example, the entrepreneur could be facing the choice among N alternative external financing options, each one having an array of possible results, and whose cost may vary depending on the conditions of the economy at the time of repayment. The entrepreneur chooses one of these actions each time, observes its consequence, and, over time, modifies his decisions on the basis of newly acquired experience. Of course his knowledge depends on the outcomes of previous decisions. The modifications he introduces in his information set show that he learns, while the information set itself represents his stock of knowledge. Thus, in an uncertain environment, we may think of learning in an iterated-choice context as the process of updating the probabilities of choosing any particular action as new information on its consequences is received.

An important aspect of our iterated-choice problem is the trade-off between the use of previously acquired knowledge and the incentive to explore a new course of action. After all, by definition, entrepreneurs are individuals who deviate from the mean and use alertness as the vehicle for creating and exploiting profit opportunities. Initially, based on instinct and existing knowledge, the entrepreneur makes some decisions and waits for their consequences. As results become available, he begins repeating only those choices that have performed better according to his initial goals. Their increased likelihood implies that actions perceived as superior exhibit increasing return to adoption, regardless of their actual superiority. This characteristic is important because it implies that, due to random events, the outcomes observed by some entrepreneurs from some specific choices may differ from the outcomes observed by other entrepreneurs when taking similar actions. This means that experiences differ across entrepreneurs thereby pushing them into following different personal patterns of choice.

Clearly, if personal patterns of choice depend on random events (luck) and on the experiences of the specific entrepreneurs, they may converge upon sub-optimal trajectories. The possibility of sub-optimal patterns is fundamental in shaping the theoretical set-up of our model. In most traditional models of learning, rational agents update their decisional algorithms in a Bayesian fashion and converge to optimal strategies (Jackson, Kalai, & Smorodinsky, 1999; Jovanovic & Nyarko, 1996). One interesting feature of our model, however, is its ability to show that entrepreneurs learn even from failure. Thus, in an alternative to strict rational formulations, our model allows entrepreneurs to make forecasting mistakes. Our argument is consistent with real-world observations, where entrepreneurs do make mistakes and do not possess complete information. For the sake of argument, however, let us assume that entrepreneurs are perfectly rational. In such an environment, two cases are possible. If the discount rate, that is the rate at which entrepreneurs discount the value of additional exploration, is set equal to zero, the optimal strategy suggests that it pays to explore new options forever. This would imply that entrepreneurs attach no value to the time and effort spent on trials. In this case, entrepreneurs converge upon the optimal action. On the other hand, if positive transaction costs limit an entrepreneur's motivation to search, then the discount rate must be positive. In this case, the optimal strategy suggests that it pays to restrict the set of possible choices down to exploit only actions that have been already tried out and have performed successfully. In some cases, these actions may happen to be objectively the one with the most desirable outcome for the entrepreneur, but, in other cases, the actions emerging as the dominant choice may be such only because early disappointing payoffs on alternative and superior choices have biased the stock of knowledge.

To summarize, with rationality and positive discounting, multiple equilibria may arise and the sequence of choices matters in determining what actions the entrepreneur will select. This is the case because the long-run outcome depends on the random consequences of earlier choices. In other words, the strategy adopted is path dependent. This shows that, in the presence of positive discounting, even when assuming rational expectations, the most likely outcomes include multiple equilibria and possibly inefficient solutions. As a result, standard path-dependence properties are shown to characterize the dynamic of the learning process even in the rational expectation formulation. Thus, to complement *ratex* formulations, and borrowing from the machine-learning literature developed in computer science (Biethalm & Nissen, 1995), we propose a learning algorithm that is mathematically tractable without requiring all typical restrictions of more traditional non-linear stochastic models.

THE MODEL

Consider an entrepreneur who, at a given time t , is ready to choose one among N possible actions, and who updates his probabilities of taking each action on the basis of the outcomes he observes from previous choices. Each action j generates an outcome $\Psi(j)$, which is unknown to the entrepreneur and randomly selected from a given distribution. Before choosing, an entrepreneur knows how he feels about deciding in favor of a specific action—that is, how confident he is that that action will produce an outcome in line with his intent. Thus, let us assume that the entrepreneur attaches a certain level of confidence, C_p , to each action, and that S_t is the sum of these levels of confidence. Let the vector p_t represents the probabilities that the entrepreneur will choose actions 1 through N . The entrepreneur learns by associating a vector of confidence levels, C_p , with each action from 1 through N .⁴

4. The model presented in this section follows significantly Arthur (1995, ch. 8).

At each time t , the entrepreneur calculates the probability vector on the basis of the relative confidence associated with each action. He then selects one action from the set according to the probabilities p_t . After the decision is made and the action taken, the entrepreneur observes the outcome of his decision and, on the basis of this new information, he updates his confidence levels by adding the outcome value of the chosen action j to action j 's pre-existing confidence level. In other words, when action j is chosen, the entrepreneur sets confidence levels to $C_t + \beta_t$, where $\beta_t = \Psi(j)u_j$, and u_j is the j th unit vector. That is, the entrepreneur updates his stock of knowledge on the basis of new experience. Finally, our entrepreneur normalizes the confidence levels to equal a pre-established constant value. That is, he sets $S_t = S$. This means that, on the basis of observed outcomes, some actions gain relative strength while others lose it.

The vector of confidence levels, C_t , can be thought of as summarizing the current confidence that the entrepreneur has learned to associate with actions 1 through N . Clearly, the confidence associated with each action changes according to the random outcome generated by that action when chosen. In other words, the entrepreneur chooses his actions with probabilities that are proportional to his current confidence in each action and learning takes place as these probabilities are updated. The decisional tree simply ensures that actions exhibiting higher payoffs, over time, become associated with higher confidence levels. Actions that pay off well are strengthened and will be taken more frequently. This means that, over time, superior actions are more likely to be chosen whereas inferior actions are more likely to be dropped. Superior, of course, is not synonymous with optimal.

In our model of choice, decisions take place randomly on the basis of current probabilities whereas the payoffs are drawn randomly from a distribution. The feature of undertaking actions randomly, on the basis of their strength, allows the model to account for the possible exploration of new options. Because of this random component, if a rarely chosen action should generate a strongly positive outcome, the level of confidence associated with it would be strengthened significantly and transform that action into a frequently chosen one.

The dynamics of entrepreneurial learning is summarized in the steps taken by the entrepreneur when his confidence levels vector is updated. At time $t + 1$, we have

$$C_{t+1} = \frac{S_{t+1}}{S_t + Y_t} (C_t + v_t) \quad (1)$$

Where C_{t+1} is the updated vector of confidence levels, S_{t+1} is the updated sum of confidence levels, and the scalar random variable Y_t is the component sum of the vector v_t .

Equation (1) may be rewritten as

$$\frac{C_{t+1}}{S_{t+1}} = \frac{C_t}{S_t + Y_t} + \frac{v_t}{S_t + Y_t} = \frac{(S_t + Y_t)C_t}{(S_t + Y_t)S_t} - \frac{Y_t C_t}{(S_t + Y_t)S_t} + \frac{v_t}{S_t + Y_t} \quad (2)$$

Each action is chosen according to its current relative confidence levels; that is, the probability of its being chosen is given by $C_t/S_t = p_t$. Substituting the random variable $[1/(S_t + Y_t)]$ with α_t , equation (2) becomes

$$p_{t+1} = p_t + \alpha_t(v_t - Y_t p_t) \quad (3)$$

Now let

$$f(p) = E[v(p) - Yp|p] \quad (4)$$

describe the conditional expectation of the change in the confidence level of action j given the action probabilities p , where the expectation is taken both over the distribution of each action's payoff and with respect to the randomly chosen actions. For any action j , triggered with probability $p(j)$, the expectation of $v - Yp$ is the vector $\psi(j)(u(j) - p)$. Thus, equation (4) can be rewritten as

$$f(p) = \sum_j \psi(j) [u(j) - p]p(j). \quad (4')$$

Equation (4') describes the process of adjustment for the distribution of probability that each action will be repeated; in other words, it is the motion vector of the algorithm. Now let us define the random vector $\varphi_t(p_t)$ as

$$\varphi_t(p_t) = v_t - Y_t p_t - f(p_t).^5 \quad (5)$$

As a result, the algorithm's dynamics described in equation (4') can be rewritten as

$$p_{t+1} = p_t + \alpha_t f(p_t) + \alpha_t \varphi_t(p_t) \quad (6)$$

Equation (6) tells us that the action probabilities are updated at each time by an "expected motion" vector $f(p_t)$ and by an unbiased "perturbation" term φ_t . In an alternative, equation (6), can be interpreted as showing that changes in the probability of choosing action j are driven by the difference between its expected payoff and the weighted-average expected payoff for all possible actions at the current probabilities p , plus some unbiased noise. The step size of the learning algorithm, α_t , is random and given by $[1/(S_t^\sigma + Y_t)]$, where the rate δ represents the speed at which the step size falls off. Thus, the model shows that the overall rate of learning increases because of both larger step size and larger differences in expected payoff among alternative possible actions. In either case, the limiting behavior of the process depends on the deterministic part of the system, that is, on equation (6) without the φ_t term.

In conclusion, do entrepreneurs really learn? In other words, do they explore enough alternative options so that, over time, they may eventually converge to selecting only the choice with highest expected payoff? The dynamics of the model enable us to analyze whether entrepreneurs tend to explore alternative choices sufficiently to be sure to concentrate, eventually, on the best available option or, instead, tend to switch patterns of action. Consistently with empirical observations, our model does not allow the a priori determination of whether or not convergence to the optimal action will prevail. On the one hand, the dynamics of the learning process show that its expected motion progresses towards the optimal payoff action. On the other hand, the learning process is also shown to be subject to positive feedback so that any high-payoff action triggered early in the process may gain reputation and become preferred by the entrepreneur, thereby locking in the dynamics. In this case, the optimal payoff action may not receive enough attention and, in the end, become obsolete. Which of these two situations will prevail depends on the rate of decrease of the step size, that is, on the value of the parameter δ . If the step size remains of constant order, an inferior action, if emphasized early, may indeed build up sufficient strength to shut out the optimal action and lock the process of choice into an inferior pattern.⁶ On the other hand, if δ is sufficiently large, this will not happen. If, for example, $\delta = 1$, the sequence $S_t = S_t'$ increases linearly, and the step size falls at the rate $1/t$. As a result, the possible lock-in to a sub-optimal action is delayed and the option to continue exploring is kept open for an arbitrarily long period of time. Even if a sub-optimal action is preferred early in the learning process, sooner or later the optimal action will be selected, tried out, and eventually preferred. Thus, whether long-run

5. Notice that f is continuous and that, by definition, the conditional expectation $E[\varphi_t/p_t]$ is zero.

6. On increasing returns and lock-in effects see Arthur (1989).

optimality is guaranteed depends on whether the value of δ is large enough to prevent premature lock-in.

It is not uncommon to observe companies generating very high returns for quite some time in spite of unsustainable or evidently shortsighted strategies. It is also difficult to convince an executive that a strategy is wrong if the returns are high. Humans tend to repeat actions and decisions that have worked in the past. Thus, in general, the stronger the payoff received from previous actions, the higher the probability that the action will be repeated, and the lower the value of δ , that is, the more likely it is that exploration of alternatives will be limited.

RATIONALITY, PATH-DEPENDENCY AND ENTREPRENEURIAL LEARNING

In our model, agents have myopic foresight. The reason for this choice is twofold. First, empirical observation shows that entrepreneurial decisions may be systematically biased and, therefore, inconsistent with a model in which agents form rational expectations. Second, as outlined earlier, we choose to model our entrepreneurs as myopic because the introduction of rationality would diminish the descriptive power of the framework while generating the same qualitative results. In addition, our conclusions are still consistent with a limiting case in which entrepreneurs may be rational. Since the purpose of the model is to describe the process of entrepreneurial learning, rather than predict its final outcome, a framework that stresses transition dynamics seems more appropriate.

A closer look at the dynamic properties of the model offers some interesting insights on the progressive development of alternative patterns of entrepreneurial choice and on their dependence on the past experiences of the entrepreneur. Our argument, after all, rests on the assumption that entrepreneurs learn from successes as well as from failures, and that their ability to succeed depends, indeed, on their capacity to process the information they acquire from experience and learn from it. Of course, good luck in the form of random chance also plays an important role.

First, our model describes entrepreneurial learning as generated, at least in part, by the reinforcement of the belief in certain actions due to their positive outcomes. This means that actions that generate initial positive outcomes acquire a self-reinforcing strength. That is, they exhibit increasing returns to adoption. Under these conditions, patterns of entrepreneurial choice are not predictable. Predictability would imply the possibility of forecasting exactly what the sequence of entrepreneurial choices will be. That is, it would permit ex-ante deterministic predictions of the potential for success of different entrepreneurs. But the sequence of actions is entrepreneur-specific and depends on random payoffs. By allowing multiple equilibria to exist as a result of increasing returns, the model shows that ex-ante knowledge of individuals' characteristics, though necessary, is not sufficient to anticipate which course of actions they will choose. This accounts for entrepreneurial creativity. In addition, since outcomes are random, knowledge of past choices of entrepreneurs is not sufficient to predict what the outcome of such choices will be and, as a result, our model can account for successes and failures as well. This feature of the model accounts for the uncertainty embedded in the entrepreneurial process.

Second, the model shows that entrepreneurial learning tends to be path-dependent. The self-reinforcing nature of positive outcomes implies that, randomly, a particular sequence of choices causes the distribution of confidence levels calculated by each entrepreneur to bend toward a preferred distribution of choices among all the possible ones. A different sequence, however, would have bent it toward an alternative distribu-

tion. Thus, each observed payoff contributes to the determination of the information set, and, therefore, it will be used, at least to some extent, in selecting future options. Clearly, early experiences push an entrepreneur toward a specific path and, if equilibrium is reached and exploration stops, later choices will be unable to push him away from that path. A different past history, however, would have put him on an alternative track and his sequence of decisions would be different.

Having established that, at least to some degree, entrepreneurial learning can be replicated using a simple repeated multi-choice problem and a reinforcement-learning algorithm, let us now return to our initial question. What can be said about the importance of path dependence and of long-run optimality for entrepreneurial learning? We know that, depending on the value for δ , learning may internalize an inferior action into the decisional process. In practice, this means that whether the optimal action dominates in the long run depends on how difficult the choice problem is. As mentioned earlier, where the current accumulation of outcomes matters, it is reasonable, both economically and psychologically, to assume that actions with higher outcomes will be chosen more and more often. Whether reinforcement leads to path dependence or not depends upon whether exploration of lower outcomes actions continues at a rate sufficient to eventually uncover the action with largest expected value. If exploitation outweighs exploration, learning may converge too rapidly on promising-looking actions. Thus, what is crucial to the emergence of the optimal action is a slowing down in speed of convergence, so that the entrepreneur has time to explore alternative actions. This speed, of course, depends on δ . Very recently, Choi and Shepherd (2000) and Fiet, Diskounov, and Gustovsson (2000) have developed theoretical frameworks on entrepreneurial knowledge and search that are consistent and complementary to ours.

In conclusion, by showing that the choices of entrepreneurs are influenced by their previous experiences, our model, thanks to its property of non-predictability and path-dependency, describes the importance of learning in shaping the long-run potential of different individuals. Furthermore, the likelihood of converging to the optimal long-run equilibrium depends on the difficulty of discriminating among the action outcomes. The reason, of course, is that, once again, where discernment of the optimal action is difficult, agents may concentrate on actions that are working reasonably well, and not explore less known but potentially superior alternatives. The existence on sub-optimal long-run equilibria allows the model to account for the possibility of failure.

CONCLUSION

The paper provides a theoretical framework describing the learning process characterizing entrepreneurial behavior. Specifically, the paper explores the connection between increasing returns to adoption and knowledge when agents choose repeatedly among actions with potentially risky consequences. When making decisions, entrepreneurs choose one of two possible strategies: In the first case they branch out in markets that are closely related to the ones in which they already operate. In the second case, entrepreneurs start new ventures in markets that are significantly separated from the ones in which they are already active. In either case, the decision taken by the entrepreneur is a function of two sets of knowledge: direct knowledge of a specific market and general knowledge of "how to be entrepreneurial." We focus on the latter and present a model in which knowledge is gained from successes as well as from failures.

Our argument is based on the idea that most learning takes place by filtering signals obtained by experimenting with different competing hypotheses, where some actions are reinforced and others weakened as new evidence is obtained. Over time, we argue, individuals repeat only those actions that have generated better outcomes. Thus, actions

exhibiting positive outcomes acquire a higher probability of being repeated, while actions with negative outcomes become less and less frequent. As a result, independently from objective desirability or actual outcomes, actions whose random outcomes happened to be positive become systematic components of the knowledge stock upon which entrepreneurs form their decisions.

Because of the self-reinforcing property acquired randomly by some actions, entrepreneurs do not necessarily follow optimal strategies. In practice, the possible lack of optimality would not have significant consequences if decisional algorithms were independent. But, most often, they are not. Earlier decisions influence significantly options available later. In addition, knowledge gained in earlier problems is used later to solve similar ones and becomes embedded in expectations and beliefs. Finally, entrepreneurs influence each other and influence the economic environment around them. "There is thus an 'ecology' of decision problems in the economy, with earlier patterns of decisions affecting subsequent decisions. This inter-linkage would tend to carry sub-optimality through from one decisions setting to another. The overall economy would then follow a path that is partly decided by chance, is history-dependent, and is less than optimal" (Arthur, 1995, p. 153).

Although this is a purely theoretical paper, empirical evidence suggests that our model does describe entrepreneurial learning. Specifically, our model suggests that entrepreneurs tend to over-exploit actions that generate initially desirable outcomes, thereby exposing themselves to the risks and benefits associated with the properties of path dependence and, in particular, the possibility of settling in an inferior pattern of choice. In many circumstances, where the superior action stands out clearly, this risk is very limited. However, the problem is likely to occur where outcomes are closely clustered, random, and therefore difficult to distinguish from each other. This is consistent with entrepreneurial behavior in highly uncertain circumstances.

In conclusion, we believe the contribution of our paper to be twofold. First, we provide a possible description of the decisional process followed by entrepreneurs. To our knowledge, no such model existed before. Second, we replace the perfectly rational agents of traditional economic models with entrepreneurs that learn from experience, process information, and, possibly, make mistakes. Throughout this process, our entrepreneurs update their decisional algorithms and, hopefully, improve their performance and face possible failures without losing their enthusiasm.

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